

Pothole Detection

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Abstract

This report presents a two-stage object detection pipeline for identifying potholes in road images. First, Selective Search generates candidate bounding boxes. Second, a ResNet-based classifier labels each proposal as pothole or background, trained on cropped proposals with labels assigned using IoU thresholds. We address class imbalance through undersampling and apply Non Maximum Suppression to reduce overlapping detections. The detector achieves 93.8% classification accuracy and 25% AP@0.5, highlighting how proposal quality and class imbalance limit detection performance. <https://github.com/thorhojhus/02516>.

1. Introduction

Potholes are depressions in asphalt pavement that arise when traffic and water damage the road surface. They reduce driving comfort and can damage vehicles, and they are a safety problem for cyclists and motorcyclists. Manual inspection of roads is time consuming and expensive, which motivates automatic vision systems that can detect potholes from images collected by vehicles or roadside cameras.

In this project we study a simplified object detection pipeline for potholes in real road scenes. Instead of training a full end-to-end detector, we follow a two stage approach. First, a region proposal method such as Selective Search [3] or Edge Boxes [4] generates a set of candidate bounding boxes for each image. Then, a convolutional neural network (CNN) classifies each proposal as either pothole or background. At test time we combine the classifier scores with Non Maximum Suppression (NMS) to obtain a final set of detections.

The project is divided into three parts that mirror a typical object detection workflow. In the first part we analyse the Potholes dataset and generate and evaluate region proposals. In the second part we train CNN based classifiers on cropped proposals while taking the strong class imbalance between background and potholes into account. In the third part we apply the trained classifier to unseen test images and evaluate the resulting detector using the Average Precision (AP) and mean Average Precision (mAP) metrics [1] at an

Intersection over Union (IoU) threshold of 0.5.

2. Method

2.1. Dataset

We use the Potholes dataset provided in the course material, which contains road images with one or more annotated potholes. Each image is accompanied by an XML annotation in Pascal VOC format that stores the image filename and one or more ground truth bounding boxes.

Following the project description we split the dataset into training, validation, and test sets with an approximate ratio of 80%/10%/10%. The split is implemented in the preprocessing script by shuffling the list of XML files with a fixed random seed and then slicing it into three disjoint subsets. This ensures reproducible experiments.

For each image we first run a region proposal method to obtain a large set of candidate bounding boxes. In our implementation we support both Edge Boxes and Selective Search and use Selective Search in the final experiments. Let B_{prop} denote a proposed box and B_{gt} a ground truth box. We assign a label to each proposal based on the Intersection over Union

$$\text{IoU}(B_{\text{prop}}, B_{\text{gt}}) = \frac{|B_{\text{prop}} \cap B_{\text{gt}}|}{|B_{\text{prop}} \cup B_{\text{gt}}|}.$$

For a given proposal we compute the maximum IoU over all ground truth boxes. If the maximum IoU is greater than or equal to a positive threshold $\tau_{\text{pos}} = 0.5$ we label the proposal as pothole (class 1). If the maximum IoU is less than or equal to a background threshold $\tau_{\text{neg}} = 0.1$ we label it as background (class 0). Proposals with IoU values in between the two thresholds are discarded since they are ambiguous.

The preprocessor saves, for each split, a JSON file that contains the image path, the list of ground truth boxes, and all labeled proposals. During training we crop each proposal from the original image and resize it to a fixed resolution of 224×224 pixels. Intensities are scaled to $[0, 1]$ and converted to PyTorch tensors. We do not apply heavy data augmentation in order to keep the focus on the proposal and detection pipeline itself.

2.2. Model Architecture

We treat proposal classification as a binary image classification problem. Let $x \in \mathbb{R}^{3 \times 224 \times 224}$ denote a cropped proposal and $y \in \{0, 1\}$ the corresponding label. The model outputs logits $f(x) \in \mathbb{R}^2$ that are converted to class probabilities with a softmax layer.

We use a ResNet-style architecture [2] consisting of a frozen ResNet-18 backbone followed by a linear layer for binary classification. This architecture has capacity to capture features that distinguish potholes from background patterns such as shadows or road markings.

2.3. Loss Functions

For training the proposal classifier we use the standard cross entropy loss for multi class classification. Given logits $f(x) \in \mathbb{R}^2$ and the ground truth label $y \in \{0, 1\}$ the loss for a single example is

$$\mathcal{L}_{\text{CE}}(x, y) = -\log p_y, \quad p = \text{softmax}(f(x)).$$

The overall training loss is the average of $\mathcal{L}_{\text{CE}}$ over all proposals in a mini batch.

The dataset is heavily imbalanced since most proposals correspond to background. To alleviate this we perform undersampling of the background class when constructing the training loader. We select all positive proposals and randomly sample a subset of negative proposals such that the fraction of positive examples in the training set is approximately one third. This simple rebalancing strategy makes the cross entropy loss more sensitive to misclassified potholes without modifying the loss function itself.

2.4. Evaluation Metrics

We use different metrics for the proposal classification task and the full object detection task.

For classification we report the accuracy on the validation and test splits, where accuracy = $\frac{\text{correct predictions}}{\text{all proposals}}$, together with the average cross entropy loss. Accuracy is easy to interpret, but it can hide the effect of class imbalance, therefore we also inspect per class confusion matrices in the experiments.

For object detection we follow the PASCAL VOC style evaluation [1]. For each image we apply the classifier to all proposals, interpret the softmax probability of the pothole class as a confidence score, and run Non Maximum Suppression (NMS) with an IoU threshold of 0.5 to remove highly overlapping boxes. A detection is considered correct if its IoU with a yet unmatched ground truth box is at least 0.5.

We sort all detections across the dataset by decreasing confidence and compute precision and recall after each detection. From the resulting precision recall curve we compute the Average Precision (AP) using 11-point interpola-

tion and report AP for the single class. We report AP rather than mean Average Precision (mAP).

2.5. Training

All experiments are implemented in PyTorch. We use the AdamW optimizer with a learning rate of $3 \cdot 10^{-4}$ and default momentum parameters. Mini batches contain 64 proposal crops. Unless otherwise stated we train each model for a small number of epochs and monitor the validation loss and accuracy after every epoch.

To reduce variance we fix the random seeds for Python, NumPy, and PyTorch, and we use deterministic shuffling of the dataset. The training script constructs the data loaders from the preprocessed JSON files, applies the undersampling strategy for the training split, and evaluates the model on the validation split after each epoch. The final model used for detection is chosen based on the validation performance.

During the detection stage we load the trained model, run it on all proposals for the validation or test images, apply NMS, and then compute AP as described above. This stage uses the same preprocessing and resizing as the training stage to ensure consistency between training and evaluation.

3. Results

3.1. Region Proposal Analysis

Table 1 summarises the output of Selective Search across all splits. On average, 677.3–745.1 proposals are generated per image, with negative proposals outnumbering positives by roughly 27.2:1. This imbalance motivates the balanced sampling strategy used during training.

Table 1. Region proposal statistics using Selective Search.

Metric	Train	Val	Test
Images	532	66	67
GT boxes	1392	187	161
Total proposals	360327	49179	40622
Positive proposals (IoU ≥ 0.5)	12784	1987	1402
Negative proposals (IoU < 0.1)	347543	47192	39220
Neg:Pos ratio	27.2:1	23.8:1	28.0:1

3.2. Proposal Classification

The ResNet classifier achieves 93.76% validation accuracy with a loss of 0.1758 (Table 2). While accuracy appears high, per-class metrics reveal the effect of class imbalance: precision is only 39.29%, but recall reaches 76.40%. The model correctly identifies 1661 of 2174 pothole proposals but produces 2567 false positives on background regions, a high false positive rate that propagates to the detection stage.

Table 2. Proposal classification results on the validation set.

Metric	Value
Validation Loss	0.1758
Validation Accuracy	93.76%
Precision (Pothole)	39.29%
Recall (Pothole)	76.40%
F1 Score	51.89%

3.3. Object Detection

Table 3 presents final detection performance. The detector achieves AP@0.5 of 24.99% with detection recall of 78.88%, localising 127 of 161 ground truth potholes at the cost of 16786 false positives. The precision-recall curve (Figure 1) shows precision reaching 60–80% at low recall but dropping sharply as recall increases.

Table 3. Object detection results on the test set (IoU threshold = 0.5).

Metric	Value
AP@0.5	24.99%
Total Ground Truth Boxes	161
True Positives	127
False Positives	16786
Detection Recall	78.88%

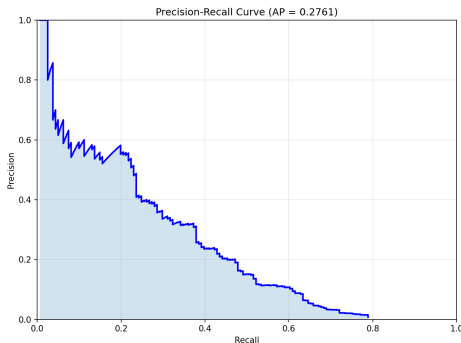


Figure 1. Precision-recall curve for pothole detection (AP = 24.99%).

Figure 2 shows a qualitative example. The ground truth (green) is matched by the highest-confidence prediction (red, score 0.92), but numerous low-confidence predictions (orange) fire on textured road regions, illustrating the primary failure mode.

4. Conclusion

The results highlight both strengths and limitations of the two-stage pipeline. The high detection recall (78.88%)

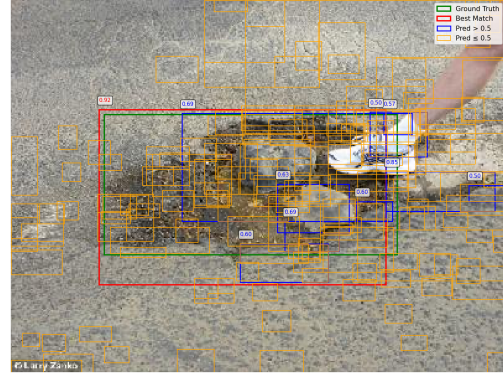


Figure 2. Detection visualisation. Green: ground truth; Red: best match; Blue: confidence > 0.5; Orange: confidence ≤ 0.5 .

demonstrates that Selective Search combined with a ResNet classifier can successfully locate most potholes. However, the low precision (39.29%) and modest AP@0.5 (24.99%) reveal two key bottlenecks.

First, the extreme class imbalance (approximately 27.2:1 negative to positive in training proposals) makes it difficult for the classifier to avoid over-predicting the pothole class. Despite undersampling of the background class during training to increase the positive fraction to about one third, the model produces many false positives (16,786) on visually ambiguous regions such as shadows, dark stains, and textured asphalt, as seen in Figure 2. Second, proposal quality imposes an upper bound on recall: small or elongated potholes may not be covered by any proposal with sufficient IoU, resulting in unavoidable false negatives.

Overall, these experiments demonstrate that a straightforward two-stage pipeline can detect potholes in challenging real-world conditions, but addressing class imbalance and improving proposal quality are essential for higher AP@0.5.

References

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